

NIRVANA – Master Specification

Volume 9: Backend & Platform Architecture

Technology Stack

Frontend: Next.js + Tailwind. Backend: FastAPI. Database: PostgreSQL. Analytics: Pandas, NumPy, OR-Tools/CVXPY. AI: Claude/OpenAI APIs.

Core Services

Authentication, User Management, Portfolio Service, Recommendation Service, Market Data Service, Credit Card Service, Tax Service, Report Service, AI Service and Admin Service.

API Design

RESTful endpoints for onboarding, portfolios, goals, reports, recommendations, uploads, Update Data, AI chat and admin operations.

Background Jobs

Scheduled market updates, provider synchronization, Excel validation, report generation, portfolio rebalancing alerts and notification delivery.

Security

JWT authentication, role-based access control, encrypted secrets, audit logging, input validation and rate limiting.

Scalability

Separate data ingestion, recommendation engine and AI services. Cache frequently accessed market data. Queue long-running tasks.

Deployment

Frontend on Vercel, backend on Railway/Render, PostgreSQL managed service, scheduled jobs for provider refresh and monitoring dashboards.